

# Jared Fernandez

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## EDUCATION

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| <b>Carnegie Mellon University: School of Computer Science</b>            | 2020 – Expected 2026 |
| Ph.D. Candidate in Computer Science, Language and Information Technology |                      |
| <i>Advisors:</i> Emma Strubell & Yonatan Bisk                            |                      |
| <b>Northwestern University, <i>Magna cum Laude</i></b>                   | 2015 – 2019          |
| B.S. in Computer Science with Honors & B.S. in Electrical Engineering    |                      |
| <i>Advisors:</i> Doug Downey & Thrastos Pappas                           |                      |

## RESEARCH EXPERIENCE

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| <b>Visiting Researcher</b>                                                            | 2024                     |
| <i>Meta, Fundamanetal AI Research (FAIR)</i>                                          | <i>New York, NY</i>      |
| <b>Visiting Researcher</b>                                                            | 2023                     |
| <i>Allen Institute for Artificial Intelligence</i>                                    | <i>Remote</i>            |
| <b>Graduate Student Researcher</b>                                                    | 2020 – Present           |
| <i>Carnegie Mellon University, Language Technologies Institute</i>                    | <i>Pittsburgh, PA</i>    |
| <b>Visiting Researcher</b>                                                            | 2022                     |
| <i>Google Research, Perception</i>                                                    | <i>Mountain View, CA</i> |
| <b>Undergraduate Researcher</b>                                                       | 2018 – 2019              |
| <i>Northwestern University, Pappas Group</i>                                          | <i>Evanston, IL</i>      |
| <b>Undergraduate Researcher</b>                                                       | 2017 – 2019              |
| <i>Northwestern University, WebSAIL Group</i>                                         | <i>Evanston, IL</i>      |
| <b>Research Intern</b>                                                                | Summer 2013, 2014        |
| <i>Stanford University: School of Earth Sciences, Cosmogenic Radiation Laboratory</i> | <i>Stanford, CA</i>      |

## INDUSTRY EXPERIENCE

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|---------------------------------------------|--------------------------|
| <b>Software Engineer</b>                    | 2019 – 2020              |
| <i>Google, Search Notifications</i>         | <i>Mountain View, CA</i> |
| <b>Software Engineering Intern</b>          | Summer 2019              |
| <i>Google, Dialogflow NLU</i>               | <i>Mountain View, CA</i> |
| <b>Software Engineering Intern</b>          | Summer 2018              |
| <i>Google, Assistant on Android Auto</i>    | <i>Mountain View, CA</i> |
| <b>Research Support Intern</b>              | Summer 2015              |
| <i>SLAC National Accelerator Laboratory</i> | <i>Menlo Park, CA</i>    |

## FELLOWSHIPS & AWARDS

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|                                                              |             |
|--------------------------------------------------------------|-------------|
| <b>NSF Graduate Research Fellowship</b>                      | 2021 – 2024 |
| <b>Carnegie Mellon University Research Fellowship</b>        | 2020 – 2021 |
| <b>Departmental Honors, Northwestern University</b>          | 2019        |
| <b>Tau Beta Pi, Northwestern University</b>                  | 2018        |
| <b>Eta Kappa Nu, Northwestern University</b>                 | 2018        |
| <b>Undergraduate Research Grant, Northwestern University</b> | 2017        |

1. **Empirically-Calibrated H100 Node Power Models for Accurate AI Training Energy Estimation.**  
 Alex C. Newkirk, **Jared Fernandez**, Jonathan Koomey, Imran Latif, Emma Strubell, Arman Shehabi,  
 Constantine Samaras  
*Environmental Research: Energy (Environ. Res: Energy)*, 2025.
2. **Hardware Scaling Trends & Diminishing Returns in Large-Scale Distributed Training.**  
**Jared Fernandez**, Luca Wehrstedt, Leonid Shamis, Mostafa Elhoushi, Kalyan Saladi, Yonatan Bisk,  
 Emma Strubell, Jacob Kahn  
*Transactions on Machine Learning Research (TMLR)*, 2025.
3. **Energy Considerations of Large Language Model Inference & Efficiency Optimizations.**  
**Jared Fernandez\***, Clara Na\*, Vashisth Tiwari\*, Yonatan Bisk, Sasha Luccioni, Emma Strubell  
*The 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.
4. **Holistically Evaluating the Environmental Impact of Creating Language Models.**  
 Jacob Morrison, Clara Na, **Jared Fernandez**, Tim Dettmers, Emma Strubell, Jesse Dodge  
*International Conference on Learning Representations (ICLR)*, 2025.
5. **Gradient Localization Improves Lifelong Pretraining of Language Models.**  
**Jared Fernandez**, Yonatan Bisk, Emma Strubell  
*Findings of the ACL: EMNLP (EMNLP Findings)*, 2024.
6. **The Framework Tax: Disparities Between Inference Efficiency in Research & Deployment.**  
**Jared Fernandez**, Jacob Kahn, Clara Na, Yonatan Bisk, Emma Strubell  
*Workshop on Efficient Systems for Foundation Models at ICML (ES-FOMO)*, 2023.  
*Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
7. **CIGLI: Conditional Image Generation from Language & Image.**  
 Xiaopeng Lu, Lynette Ng, **Jared Fernandez**, Hao Zhu  
*Workshop on Closing the Loop Between Vision and Language at ICCV (CLVL)*, 2021.
8. **Generative Data Augmentation for Commonsense Reasoning.**  
 Yiben Yang, Chaitanya Malaviya, **Jared Fernandez**, Swabha Swayamdipta, Ronan Le Bras, Ji-Ping Wang,  
 Chandra Bhagavatula, Yejin Choi and Doug Downey  
*Findings of the ACL: EMNLP (EMNLP Findings)*, 2020.
9. **CODAH: An Adversarially Authored Question-Answer Dataset for Common Sense.**  
 Michael Chen, Mike D’Arcy, Alisa Liu, **Jared Fernandez**, and Doug Downey  
*Workshop on Evaluating Vector Space Representations for NLP (RepEval)*, 2019.
10. **Sampling Informative Training Data for RNN Language Models.**  
**Jared Fernandez** and Doug Downey.  
*Association of Computational Linguistics: Student Research Workshop (ACL-SRW)*, 2018.
11. **VecShare: A Framework for Sharing Word Representation Vectors.**  
**Jared Fernandez**, Zhaocheng Yu, and Doug Downey  
*Empirical Methods in Natural Language Processing (EMNLP)*, 2017.

## NON-ARCHIVAL WORKSHOPS AND PREPRINTS

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1. **Practical Evaluation of Machine Learning Efficiency Requires Model Life Cycle Assessment.**  
**Jared Fernandez**, Clara Na, Yonatan Bisk, Constantine Samaras, Emma Strubell  
*Workshop on Socially Responsible Language Modelling Research at the Conference on Language Modeling (SoLaR)*, 2025.
2. **Evaluating the Environmental Impact of Language Models with Life Cycle Assessment.**  
**Jared Fernandez\***, Clara Na\*, Yonatan Bisk, Emma Strubell  
*Climate Change AI Workshop (CCAI at ICLR)*, 2025. **Best Paper: Proposals Track.**

### 3. Efficiency Pentathlon: A Standardized Arena for Efficiency Evaluation.

Hao Peng, Qingqing Cao, Jesse Dodge, Matthew E. Peters, **Jared Fernandez**, Tom Sherborne, Kyle Lo, Sam Skjonsberg, Emma Strubell, Darrell Plessas, Iz Beltagy, Evan Pete Walsh, Noah A. Smith, Hannaneh Hajishirzi  
*Preprint (Arxiv), 2023.*

### 4. Adapting to Gradual Distribution Shifts with Continual Weight Averaging.

**Jared Fernandez**, Saujas Vadaguru, Sanket Vaibhav Mehta, Yonatan Bisk, Emma Strubell  
*Workshop on High-Dimensional Learning Dynamics at ICML (HiDL), 2023.*

## INVITED TALKS

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### 1. Energy Considerations in Efficient Language Model Development and Deployment

*Seagate Technology*  
*Bosch Research*

October 2025  
June 2025

### 2. Sampling Informative Training Data for RNN Language Models

*Midwest Speech and Language Days*

May 2018

## TEACHING EXPERIENCE

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**On-Device Machine Learning** with Profs. Yonatan Bisk & Emma Strubell

Fall 2023, 2024

**Machine Learning** with Prof. Doug Downey

Spring 2018

**Discrete Math** with Prof. Aravindan Vijayaraghavan

Fall 2017

**Artificial Intelligence** with Prof. Michael Rubenstein

Spring 2017

## MENTORSHIP

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**Xiang Fan**, *CMU BS → PhD at University of Washington*

2022 - 2023

**Vashisth Tiwari**, *CMU MS → PhD at CMU*

2025

**Mohith Rajesh**, *CMU MS*

Current

**Raj Maheshwari**, *CMU MS*

Current

## DEPARTMENTAL SERVICE

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**Graduate Application Support Program Mentor**, *Carnegie Mellon University*

2020 – 2022

**Undergraduate & Graduate Student Mentor**, *Carnegie Mellon University*

2020 – 2022

**Undergraduate Mentor**, *Northwestern Eta Kappa Nu*

2019

**External Vice President**, *Northwestern IEEE Student Chapter*

2018

**Technical Program Director**, *Northwestern IEEE Student Chapter*

2017

## PROFESSIONAL SERVICE

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### Reviewing

ACL Rolling Review

2024, 2025

NeurIPS

2022, 2024, 2025

ICLR

2022, 2024

ICML

2023

ACM Multimedia

2022

### Student Volunteer

EMNLP

2017

ACL

2018